

CSC360 - Computer Graphics and Digital Image Processing

1/9/26 Reflection

Topic: Linear Equations, Vectors, Geometry and Project Development

In today's lecture, we covered many math topics that are applicable to computer graphics and some programming concepts related to project development. The lesson was about connecting equations, vectors, matrices, and geometry to represent graphical objects.

Important Things Learnt

→ Linear Equations and Vectors:

Graphical problems can be expressed in terms of linear equations and vectors. For instance, equations like $2x + 3y = 5$ can be solved for their needed values of coordinates.

- $A \times x = b$ is a system of linear equations.
- A vector can be expressed as (x, y) .
- Vectors may be either row vectors or column vectors.
- Various systems may have distinct, infinite or no solutions at all.

Matrices

In addition, we studied matrices and their properties in detail, focusing on how to solve equations using them and test vector relationships.

- Singularity and full rank of matrices.
- Rank of a matrix.
- Testing vector dependencies.
- $n \times n$ matrix with an $n \times 1$ vector yields an $n \times 1$ resultant vector.

These principles prove very helpful in making transformations and calculations in computer graphics.

Geometry and Distance

The professor came up with an interesting project idea where three equations would be entered as text in order to form a triangle. It was also introduced to us how formulas in mathematics can be applied to find out if a certain point is within a circle or not.

- Distance between two points.
- Point relative to a circle.
- Derivation of the distance formula.

This is useful in creating and recognizing graphics objects.

Java Project Development

We have also talked about some of the programming notions that are necessary for our projects, such as Maven libraries and methods and basic data structures.

- The ability to comprehend what libraries are to be included into Maven
- The ability to know what methods are to be called
- Stack => LIFO (Last In, First Out)
- Queue => FIFO (First In, First Out)
- Event listeners are utilized in order to react to user actions within graphical applications.

Key Takeaway

Today's lesson has made it possible for me to learn how mathematics and programming are connected in computer graphics. For instance, vectors, matrices, equations, and distances can be used directly for representing and working with graphical objects, while Java and Maven are the means for this.